

ResBi-Mamba Plus: Deep Bidirectional Mamba with Spatiotemporal Attention for Robust Interpretable Hourly Runoff Forecasting

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ABSTRACT

As runoff forecasting shifts from single-step to multi-step, stakeholders in the hydrological field have placed higher demands on the multi-lead-time forecasting performance of models. In this paper, we propose a novel interpretable fine-grained runoff forecasting framework: Residual Bidirectional Mamba Plus (ResBi-Mamba Plus). The backbone of the proposed model is our redesigned bidirectional Mamba (Bi-Mamba) architecture that captures more contextual information through parallelized forward and backward sequence traversals. For multivariate long-sequence inputs, we further integrate a dual-path spatiotemporal attention (STA) module after each Bi-Mamba branch. Based on the channel-dependence (CD) and channel-independence (CI) strategies, the STA module adaptively recalibrates the original features in the temporal and spatial dimensions, so that the important information in each unidirectional branch can be highlighted during the branch fusion process. In addition, the overall framework adopts the ResNet Plus architecture, which improves information flow in the network through two parallel residual block paths and dense shortcut connections. This design allows the model to obtain clear benefits from increased depth that are difficult for conventional models to achieve. Ablation and comparative experiments on the hourly runoff dataset from the Columbia River in the United States fully verify the effectiveness of the components in ResBi-Mamba Plus. The results also show that the proposed model performs better than current state-of-the-art time series forecasting models across various forecasting horizons.

River runoff is a vital component of the natural water cycle. It not only supports the diversity and balance of river basin ecosystems, but also provides essential freshwater resources for agricultural irrigation, industrial production, and domestic use in riverside communities [1]. In addition, river runoff is the basis of hydropower

generation. Hydropower is currently the most widely used source of renewable energy and accounts for half of global low-carbon electricity generation [2]. Therefore, the scientific utilization and management of river runoff are important for both ecological protection and socioeconomic development. Runoff forecasting, which involves knowledge from mul-

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multiple disciplines, can provide reliable decision-making support for hydrology and has become a current research hotspot [3], [4].

In order to adapt to different specific tasks and scenarios, runoff forecasting is further classified into hourly forecasting [5], daily forecasting [6], monthly forecasting, and annual forecasting according to temporal resolution [7], [8]. Monthly and annual runoff forecasting are mainly used for climate change studies and long-term development planning of river basins, while daily runoff forecasting is used for the daily scheduling and maintenance of hydropower plants. Among all subcategories of runoff forecasting, hourly runoff forecasting focuses on the shortest time scale, characterized by significant randomness and uncertainty due to the influence of numerous external factors. This makes hourly runoff a non-stationary time series and the most challenging to forecast. However, hourly runoff forecasting is crucial for providing timely warnings and responses to imminent hydrological events, thus holding significant research value [9].

As a significant branch of time series forecasting, runoff forecasting has undergone decades of development and refinement, evolving into a mature and comprehensive system. Process-based models (also known as physics-based models) were among the earliest methods to be applied [10]. These models utilize data on multiple external factors affecting runoff, such as precipitation, watershed topography, soil and vegetation, to model the highly complex physical processes of the hydrological cycle [11]. A representative example is the Soil and Water Assessment Tool (SWAT) model introduced by the United States Department of Agriculture (USDA) in the 1990s [12]. This distributed model is capable of simulating various physical processes, including surface runoff, subsurface flow, and nutrient transformations within watersheds, and remains a classic model still in use within the hydrological field today. The advantages of process-driven models include their great interpretability and lower computational costs. However, their modeling process is highly dependent on specialized knowledge, making development challenging and creating a high barrier to entry. In recent years, data-driven models represented by artificial neural networks (ANNs) have become the mainstream runoff forecasting methods [13], [14], which have strong nonlinear fitting capabilities and are therefore suitable for modeling complex and abstract time series relationships in non-stationary short-term runoff sequences. Data-driven models identify relationships between inputs and outputs through the learning of a large amount of data; thus, their modeling does not rely on an understanding of complex physical processes and can be rapidly and flexibly generalized to various specific scenarios [15], [16]. With the rapid development of deep learning, traditional ANNs

have gradually become baseline models. As the performance of conventional ANNs is limited for increasingly demanding runoff forecasting tasks, several variants designed for time series forecasting have been applied in this field. The long short-term memory (LSTM) network [17], [18] is considered as the most widely used model for time series forecasting. LSTM is developed from the recurrent neural network (RNN) [19] and introduces gating units to retain useful historical information. This structure reduces the effects of vanishing gradient and improves the modeling of long-term dependencies. Han et al. [20] propose AT-LSTM for monthly runoff forecasting by combining LSTM with attention mechanisms. The proposed hybrid model assigns different weights to the elements and gives more attention to information related to the task. Gao et al. [21] further present an hourly runoff forecasting model based on the gated recurrent unit (GRU) [22]. GRU simplifies the gating structure of LSTM while maintaining similar forecasting performance, thereby reducing the number of parameters. However, LSTM and GRU are both based on the recurrent architecture. At each time step, the current calculation depends on the output from the previous step. As a result, they cannot process an input sequence in parallel. To address this problem, Lin et al. [23] apply a temporal convolutional network (TCN) [24] to runoff forecasting. As a variant of the convolutional neural network (CNN) [25], the TCN not only inherits the capability for parallel processing but also enhances the performance in time series forecasting through the use of dilated convolution and causal convolution, which expand the receptive field and prevent information leakage. Vaswani et al. [26] propose the Transformer, a seminal natural language processing (NLP) architecture which utilizes the self-attention mechanism to directly calculate dependencies between any two elements in the sequence. The Transformer combines the capabilities of parallel processing with effective capture of long-term dependencies, and has become the foundational architecture for multiple large-scale language models (LLMs) [27], including the famous Generative Pre-trained Transformer (GPT) [28]. Building on this, Yin et al. [29] present the RR-Former, the first application of the Transformer architecture to runoff forecasting, demonstrating its superior accuracy compared to the traditional LSTM model.

Although the Transformer has excellent time series forecasting capability, its self-attention mechanism calculates the relations between all pairs of elements in the input sequence. As a result, its computational complexity increases quadratically with sequence length. The quadratic complexity of the Transformer significantly increases the computational cost and limits its use for long-term forecasting. LSTM and Transformer also face

difficulties when deeper network structures are used. For LSTM, the recurrent architecture restricts its training efficiency. Meanwhile, the computational cost of the Transformer increases rapidly with sequence length. Moreover, deeper networks may suffer from performance degradation. These problems limit the extraction of high-level features from long input sequences. In hourly runoff forecasting, their effects become more evident as the forecasting horizon increases. Model accuracy often decreases at longer horizons, making it difficult to maintain stable performance across multiple lead times. Recently, Gu et al. [30] proposed Mamba, a sequence modeling architecture with linear computational complexity [31]. Mamba combines convolutional and recurrent properties and has therefore been considered as an alternative to Transformer for long-term forecasting. Although Mamba shows strong potential for long-sequence modeling, its capability in fine-grained multivariate long-term forecasting can be further improved through more effective architectural design. Motivated by these considerations, we innovatively propose a powerful hourly runoff forecasting framework: Residual Bidirectional Mamba Plus (ResBi-Mamba Plus), which deeply integrates redesigned bidirectional Mamba (Bi-Mamba), improved deep residual network (ResNet Plus), and lightweight spatiotemporal attention mechanism (STA), capable of robust and interpretable multi-lead-time runoff forecasting. The main contributions of this paper are as follows:

- 1) A novel Bi-Mamba architecture based on the modified Mamba-2 is proposed. It processes the forward and backward information flows in parallel and serves as the backbone of the proposed framework. Each unidirectional branch provides better forecasting performance than Mamba-1. By combining information from both directions, Bi-Mamba also improves long-sequence forecasting.

- 2) A lightweight attention module, STA, is added to each unidirectional branch of Bi-Mamba. It models the relations among different time steps and features in multivariate time series along both the temporal and spatial dimensions. We also adopt ResNet Plus in the overall network architecture by introducing an additional residual block pathway. Dense shortcut connections are built between residual blocks on the two pathways to optimize the forward and backward information flow.

- 3) This paper is the first to explore the potential applications of the Mamba architecture for fine-grained hourly runoff forecasting. Ablation and comparative experiments are conducted to examine the effectiveness of each component and compare the forecasting performance of the proposed framework with that of existing mainstream models.

In the proposed framework, Bi-Mamba processes the historical sequence in both forward and backward directions. STA assigns weights to temporal

and spatial features before the outputs of the two branches are fused. ResNet Plus builds the residual structure required for stacking deeper Bi-Mamba blocks. Together, these components support multi-lead-time hourly runoff forecasting with improved accuracy and interpretable feature weighting.

1 Related Work

Neural network models for time series forecasting have developed along several technical directions. According to their corresponding characteristics, the mainstream models can be divided into three categories: recurrent and convolutional models, Transformer-based models, and Mamba-based models.

The first category includes traditional time series forecasting models, such as RNNs and TCNs. Recurrent models process sequence information step by step. LSTMs and GRUs use gating mechanisms to retain useful information and remove less relevant information. However, recurrent computation makes parallel processing difficult and reduces computational efficiency. Although the TCN supports parallel processing, its small receptive field limits its ability to capture long-term dependencies.

The second category involves Transformer-based models, such as the Informer [32] and iTransformer [33]. These models use self-attention to calculate the relations between sequence elements. This structure is suitable for learning long-term dependencies and has been widely used in time series forecasting. However, the computational cost of self-attention increases quadratically with the sequence length. Therefore, Transformer-based models may require a high computational cost when handling long input sequences, making it difficult to achieve a balance between accuracy and efficiency.

The third category includes models based on Mamba. Mamba has linear computational complexity and supports efficient processing of long sequences. It introduces a selective mechanism to control how input information affects the hidden states. It also uses hardware-aware algorithms, including parallel associative scan and memory recomputation, to reduce the computational cost. These features make Mamba suitable for long-term forecasting tasks. Several variants, including Vision Mamba [34], have also been developed for other types of data and tasks. However, for multivariate long-term forecasting, the original Mamba architecture does not directly provide an attention-based mechanism to identify important temporal and cross-variable information. In this paper, we propose ResBi-Mamba Plus for hourly runoff forecasting. The proposed framework combines bidirectional branch fusion, spatiotemporal attention, and a deep residual structure to improve the forecasting of multivariate hourly runoff sequences.

2 ResBi-Mamba Plus

2.1 Problem Statement

Given historical multivariate hourly runoff series with a fixed-length look-back window containing L samples $\mathbf{X} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_L\}$, where each data point $\mathbf{x}_t \in \mathbb{R}^M$ at the t -th time step contains M variables, the objective of the model is to predict the runoff sequence $\mathcal{F} = (\mathbf{x}_{L+1}, \mathbf{x}_{L+2}, \dots, \mathbf{x}_{L+T})$ in the subsequent T hours. The forecasting process is conducted through the sliding window, whereby after each forecasting of the future runoff sequence $\mathcal{F}$ is completed, the look-back window slides forward according to a predefined step size T for subsequent forecasts. When sufficient data are available, to ensure that the predicted values closely approximate the ground truth, the model should be fed with historical data long enough to learn periodic patterns. This means the length of historical data L should be greater than or equal to the forecasting horizon T .

2.2 Overall Framework

Fig.1 shows the overall architecture of the proposed ResBi-Mamba Plus for multivariate hourly runoff forecasting. The model is designed to use deeper network structures and longer input sequences than conventional forecasting models. First, ResNet Plus is used as the main network structure. It contains two parallel residual block paths, with dense shortcut connections between the residual blocks on the two paths. These connections improve information transmission across layers and reduce the training difficulty caused by increasing network depth. Each residual block uses Bi-Mamba as its sequence modeling component. Bi-Mamba adds a backward branch to the forward branch and processes the two directions in parallel. The model can therefore use information from both directions in the input sequence. In addition, an STA module is

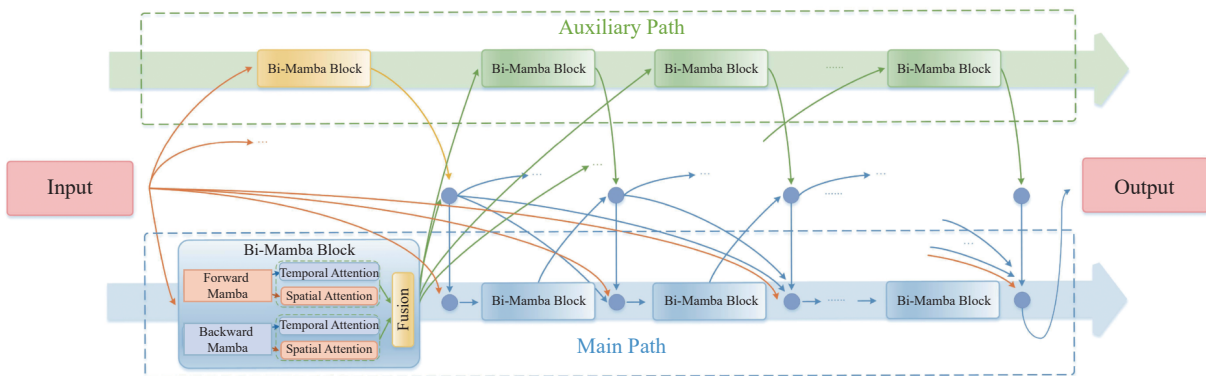
placed after each unidirectional branch of Bi-Mamba. The module assigns different importance to time steps and variables in the two branches before their outputs are fused. This helps the model retain information that is more useful for runoff forecasting. By combining these features, the proposed framework can extract features through a deeper network while maintaining robust forecasting performance across different lead times.

2.3 ResNet Plus with Denser Shortcuts and Dual Pathways

Neural networks model features through hierarchical learning. The features extracted in earlier layers are gradually transformed into more complex representations in deeper layers. In principle, increasing the network depth can improve its ability to learn complex patterns. However, deeper networks have been found to be more difficult to train in practice. As the number of layers increases, vanishing gradients and network degradation may occur, and the forecasting performance may decrease rather than improve. The ResNet proposed by He et al. [35] is considered an effective way to reduce these problems. It introduces shortcut connections that allow information to bypass one or more layers. These connections make information and gradients easier to transmit through the network. The backpropagation process of ResNet can be formalized as follows:

$$\begin{aligned} \frac{\partial loss}{\partial \mathbf{x}_l} &= \frac{\partial loss}{\partial \mathbf{x}_L} \cdot \frac{\partial \mathbf{x}_L}{\partial \mathbf{x}_l} \\ &= \frac{\partial loss}{\partial \mathbf{x}_L} \cdot \left(\mathbf{1} + \frac{\partial}{\partial \mathbf{x}_L} \sum_{i=l}^{L-1} \mathcal{F}(\mathbf{x}_i, \mathbf{W}_i) \right) \end{aligned} \quad (1)$$

where the bolded constant term ‘1’ in the equation represents the shortcut connection, which provides an additional direct path for gradient flow without any weight multiplication. This ensures the preservation of effective gradients even in very deep



Note: The model is composed of residual blocks that integrate Bi-Mamba and STA modules as fundamental building units. These blocks are stacked and interact along two parallel residual pathways, enhancing information flow throughout the network and enabling the construction of deep Mamba-based models.

Fig. 1 The Overall Architecture of the Proposed ResBi-Mamba Plus Network

layers. Furthermore, Orhan et al. [36] prove that shortcut connections can eliminate singularities caused by factors such as permutation symmetry and linear dependencies among nodes, thereby enhancing the efficiency of model training.

Based on the advantages of shortcut connections, ResBi-Mamba Plus uses ResNet Plus [37] as its overall network architecture to build a deeper Mamba-based model. ResNet Plus combines the characteristics of DenseNet [38] and Merge-and-Run mapping [39]. As shown in Fig.1, it contains two parallel residual block paths and uses dense shortcut connections between different layers. The first residual blocks on the two paths receive the same original input. From the second layer onwards, each residual block on the auxiliary path receives the output of the first residual block on the main path. At each layer, the outputs of the two residual blocks are averaged. The averaged output is then connected to the subsequent residual blocks on the main path through dense shortcut connections. Therefore, the input of each residual block on the main path contains information from the original input and the outputs of the preceding layers. This structure allows features extracted at different levels to be reused in later layers. Compared with the ResNet or Inception networks using the same number of residual blocks, ResNet Plus provides shorter paths for gradient propagation. This structure reduces the difficulty of training a deep network. In ResBi-Mamba Plus, it enables more Mamba-based residual blocks to be stacked for modeling multivariate long time series.

2.4 Bidirectional Mamba-2

Mamba is a novel sequence modeling architecture developed after the RNN and Transformer-based models. It achieves a balance between forecasting performance and computational cost when processing long input sequences. Mamba is built on the State Space Model (SSM) [40]. SSM was originally developed in modern control theory to describe the change in a system state over time. In SSM, the input $\mathbf{x}_t$ is mapped to the output $\mathbf{y}_t$ through a hidden state $\mathbf{h}_t$. This process is described by first-order differential equations as follows:

$$\mathbf{h}(t)' = \mathbf{A}\mathbf{h}(t) + \mathbf{B}\mathbf{x}(t) \quad (2)$$

$$\mathbf{y}(t) = \mathbf{C}\mathbf{h}(t) \quad (3)$$

where $\mathbf{A} \in \mathbb{R}^{N \times N}$, $\mathbf{B} \in \mathbb{R}^{N \times D}$ and $\mathbf{C} \in \mathbb{R}^{D \times N}$ represent three learnable parameter matrices, corresponding to the state transition matrix, input matrix, and output matrix, respectively. N denotes the state dimension, while D represents the variable dimension. The original SSM was designed to deal with continuous functions. However, to model the real-world discrete time series, it is necessary to transform the continuous parameters $(\Delta, \mathbf{A}, \mathbf{B})$ into discrete parameters

$(\bar{\mathbf{A}}, \bar{\mathbf{B}})$ using the zero-order hold (ZOH) method, where Δ is a special time scale parameter used to control the updating of the state at each time step. The discretization process can be expressed as:

$$\bar{\mathbf{A}} = \exp(\Delta\mathbf{A}) \quad (4)$$

$$\bar{\mathbf{B}} = (\Delta\mathbf{A})^{-1}(\exp(\Delta\mathbf{A}) - \mathbf{I}) \cdot \Delta\mathbf{B} \quad (5)$$

The formula for the discretized SSM is then given by:

$$\mathbf{h}_k = \bar{\mathbf{A}}\mathbf{h}_{k-1} + \bar{\mathbf{B}}\mathbf{x}_k \quad (6)$$

$$\mathbf{y}_k = \mathbf{C}\mathbf{h}_k \quad (7)$$

where k represents the discrete time step. The discretized SSM exhibits unique dual attributes of convolution and recursion. During the training phase, it employs convolutional operation to compute in parallel, enhancing training efficiency. In the inference phase, outputs can be rapidly generated through recursion. However, SSM still suffers from the long-term dependency problem. To address this, Gu et al. [41] replace the randomly initialized matrix $\mathbf{A}$ with the HiPPO matrix, significantly improving the SSM's retention of historical information. This development laid the foundation for the precursor to Mamba: the structured state space model (S4). Building upon S4, Mamba introduces two significant improvements: a data-driven selection mechanism and a new hardware-aware parallel algorithm. The S4 is fundamentally a linear time-invariant (LTI) model, meaning that Δ , $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$ remain constant at all time steps; thus, the model lacks context-aware ability specific to given inputs. To address this, Mamba implements a unique selection mechanism that parameterizes Δ and matrices $\mathbf{B}$ and $\mathbf{C}$ according to the input, transforming S4 into a data-driven, time-varying model. This process can be described as follows:

$$\mathbf{B} \rightarrow S_{\mathbf{B}}(\mathbf{x}) = \text{Linear}_N(\mathbf{x}) \quad (8)$$

$$\mathbf{C} \rightarrow S_{\mathbf{C}}(\mathbf{x}) = \text{Linear}_N(\mathbf{x}) \quad (9)$$

$$\Delta \rightarrow S_{\Delta}(\mathbf{x}) = \tau_{\Delta} \cdot \text{Broadcast}_D(\text{Linear}_1(\mathbf{x})) \quad (10)$$

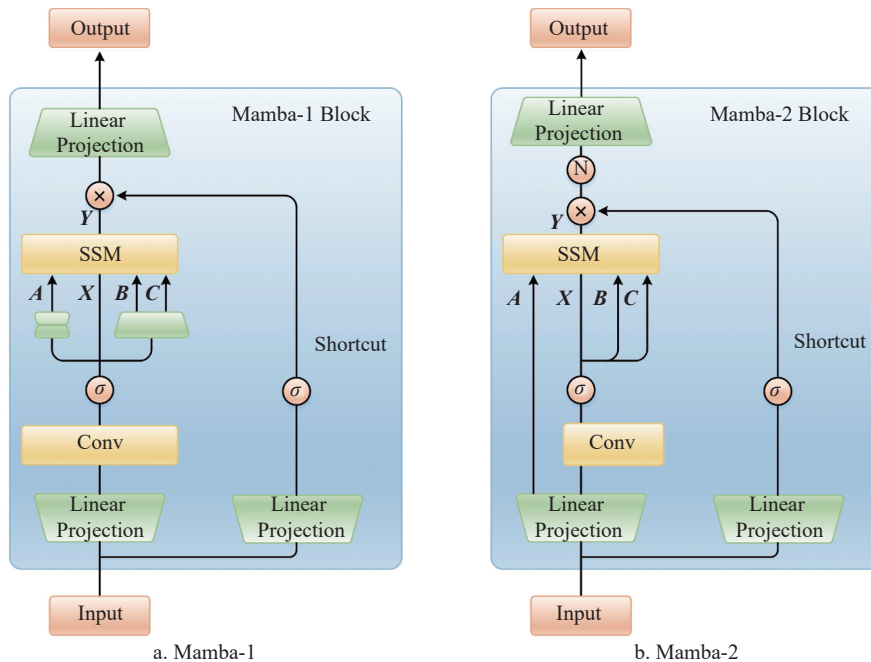
where Linear_N represents a parameterized projection to dimension N , and τ_{Δ} denotes the Softplus function. The selection mechanism in Mamba is similar to a combination of attention and gating mechanisms, enabling selective memory and forgetting of inputs and their projections. The interaction between Δ and $\mathbf{A}$ affects the information stored in the hidden state, while $\mathbf{B}$ and $\mathbf{C}$ allow the model to select input-dependent information. However, this time-varying mechanism means that Mamba can no

longer be computed as a standard convolution. Mamba therefore uses a recurrent scan algorithm instead of the convolution operation. The input sequence is divided into chunks, and the results are combined through a tree-based scan procedure. This allows the computations to be performed in parallel on modern hardware. Mamba also performs discretization and recurrent computation in GPU SRAM to reduce data transfer between different memory levels. During backpropagation, intermediate states are recomputed rather than stored throughout the forward process, which reduces memory use when long sequences are processed. The Mamba block is the basic component of the model, and its structure is shown in Fig. 2a. In its simplified form, the Mamba block combines the H3 structure based on SSM with a gated multi-layer perceptron (MLP). Multiple Mamba blocks can then be stacked in the network in a similar way to residual blocks.

Dao et al. [42] propose Mamba-2 on the basis of Structured State Space Duality (SSD). SSD establishes a mathematical connection between SSM and attention by representing both operations in the form of semiseparable matrices. This formulation allows the computation of SSM to be implemented through structured matrix multiplication. In Mamba-2, the SSD layer divides the matrix into blocks to make better use of modern accelerators. Computation within diagonal blocks is performed using a dual-attention form, while the low-rank off-diagonal blocks are used to calculate the relations between different blocks. With this matrix-based implementation, Mamba-2 can use tensor cores for SSM

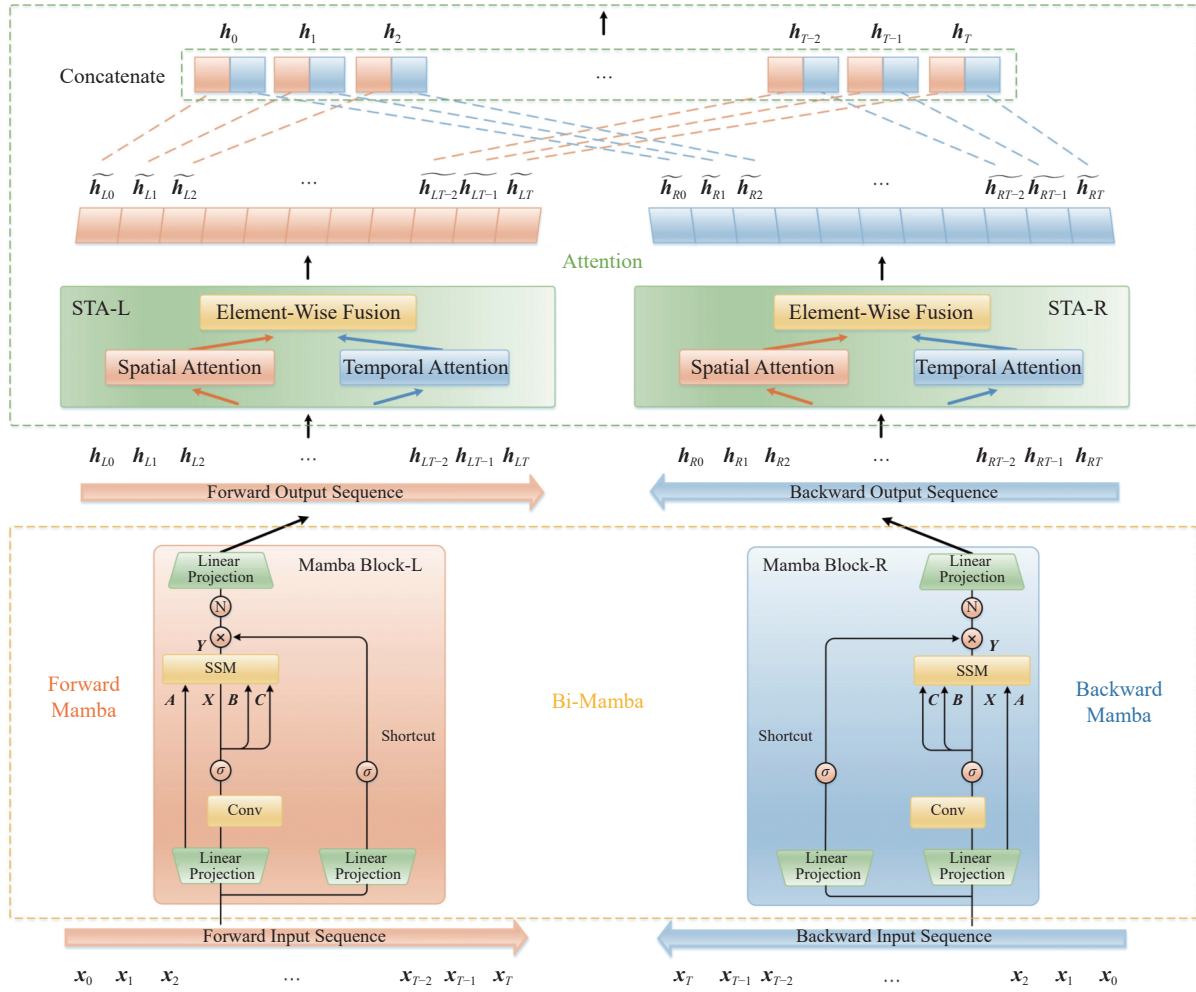
computation and achieves a reported training speedup of 2-8 times. In addition, as shown in Fig. 2b, the architecture of the Mamba-2 block is also optimized by replacing the SSM-centric sequential linear projection with a parallel parametric projection. In this configuration, (A, B, C) are no longer considered auxiliary parameters, but rather the SSD layer acts as a map from (A, X, B, C) to Y , which allows for better applicability of tensor-parallelism to the input projection. Moreover, Mamba-2 incorporates an extra normalization layer before the final output projection to ensure training stability. Given these advantages, our proposed ResBi-Mamba Plus employs Mamba-2 as its core backbone to construct the overall model.

Mamba-2 provides an efficient way to process long input sequences, making it suitable for forecasting tasks with longer lead times. In such tasks, a longer historical input may provide additional information for forecasting. However, existing Mamba models usually process this input in a single direction, from earlier observations to later ones. This structure may not fully capture the relations among observations within the historical window. Previous studies on LSTM- and SSM-based models have shown that bidirectional structures can improve forecasting performance in several tasks [43], [44]. Based on these findings, we develop Bi-Mamba using Mamba-2 as the backbone, as shown in Fig. 3. In each Bi-Mamba block, an additional Mamba branch processes the input sequence in the reverse direction, from x_t to x_0 . This backward branch only uses the historical input sequence available at the forecasting time. It does not introduce future



Note: The green trapezoid denotes linear projection, the yellow rectangle represents sequence transformations (Conv stands for convolution calculation), and the orange dot represents nonlinearity.

Fig. 2 The Internal Architectures of the Mamba-1 and Mamba-2 Blocks



Note: This architecture consists of two parallel branches: Forward Mamba and Backward Mamba. These branches traverse the input sequence in opposite directions, and their outputs are adaptively recalibrated through the subsequently embedded STA module.

Fig. 3 The Proposed Bi-Mamba Architecture

observations during either training or inference. The forward and backward branches are computed independently and can be processed in parallel. Since both branches retain the linear complexity and hardware-efficient computation of Mamba, the bidirectional structure improves sequence representation without introducing the high cost associated with quadratic-complexity models. Bi-Mamba can therefore be stacked in multiple layers in the same way as standard Mamba blocks.

Existing models based on bidirectional architecture typically perform simple element-wise concatenation when merging two independent branches. This operation does not explicitly distinguish the contributions of the forward and backward branches at different time steps and across different variables. To address this problem, we introduce a dual-path STA module. The module applies both channel-dependent (CD) and channel-independent (CI) strategies to model temporal and cross-variable relations in the two branches. Before branch fusion, the proposed STA module assigns different weights to the features produced by each branch, so that information more relevant to runoff forecasting

receives greater attention. This design further improves ResBi-Mamba Plus's temporal robustness when modeling long sequences.

2.5 Dual-Path STA Module

Time series forecasting models usually take historical information that is multiple times the length of the forecasting horizon as input to ensure the sequence contains sufficient contextual information. To address the challenges posed by extended sequences, we further integrate the attention mechanism into each Bi-Mamba block [45], enabling the model to dynamically focus on important parts of the input feature map. Previous work on Mamba-2 has also suggested that combining Mamba with attention is a useful direction for further model development. In the proposed architecture, the Mamba layer first extracts sequence representations through the SSM, and the attention module then adjusts these representations according to the importance of different time steps and variables. In this way, the two components play different roles in the forecasting process: Mamba provides efficient long-term modeling, while attention helps distinguish useful

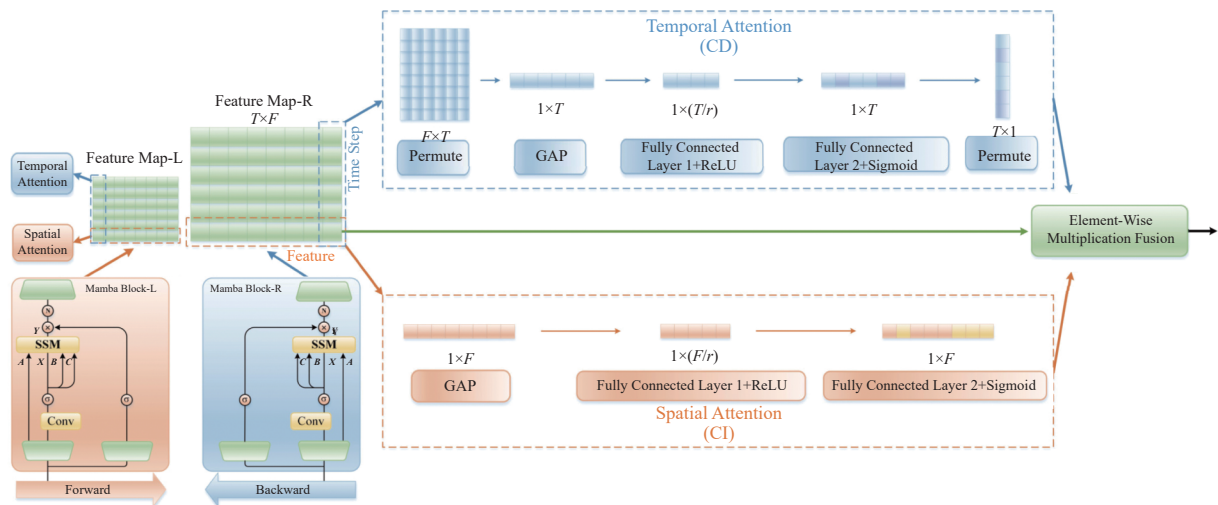
information in the extracted features. Dao et al. [42] have also demonstrated through empirical studies that combining attention with SSM significantly improves the forecasting performance of the model compared to the vanilla Mamba model. In real-world applications, time series data often contain multiple variables. Consequently, in multivariate time series forecasting, original sequences are considered as multi-channel signals where each channel represents an independently recorded univariate series. Mainstream attention mechanisms employed in time series forecasting models, such as the self-attention of Transformer, mainly focus on a single dimension, usually the temporal dimension. Under the CD strategy, information from different variables is combined at each time step to form temporal features. This allows the model to capture relations among variables, but the individual temporal pattern of each variable may not be fully preserved. This limitation may affect forecasting in hydrological and meteorological tasks, where different variables may demonstrate distinct long-term changes. Emerging models like iTransformer have progressively adopted the CI strategy. Using this strategy, the complete sequence of each variable is treated as an individual input unit, allowing the model to better represent the temporal pattern of each variable. However, CI does not directly model the relations among different variables. This can become a limitation when the forecasting task involves many related input channels.

To achieve robust long-term forecasting, models must simultaneously consider the interdependencies between features in both temporal and spatial (variable channel) dimensions. Based on this consideration, we design a lightweight STA module, as shown in Fig. 4. STA can be inserted into different parts of a time series forecasting model. In ResBi-Mamba Plus, it is placed after each

unidirectional branch of Bi-Mamba to adjust the output feature maps before the two branches are fused. The forward and backward branches may provide different useful information for runoff forecasting. STA enhances the relevant information from each branch and suppresses less useful features during the fusion process. Unlike attention modules that operate only along one dimension, STA processes the feature maps in both the temporal and spatial dimensions. In the temporal path, the CD strategy is used to capture temporal relations in the combined multivariate features. In the spatial path, the CI strategy is used to evaluate the contribution of each variable from its complete temporal sequence.

In each dimension, STA extends the successful squeeze-and-excitation network (SENet) [46] from computer vision (CV) to time series forecasting for adaptively explicit modeling. The input to STA is a feature map with dimensions $[t \times f]$ from each unidirectional Mamba output, where t and f represent the number of time steps and variable channels, respectively. The modeling process of STA can be divided into two stages: squeeze and excitation. Initially, the squeeze operation uses a global average pooling (GAP) layer along the respective dimension to compress the original two-dimensional feature map into a one-dimensional vector of size $[1 \times t]$ or $[1 \times f]$. Here, features at each time step or each independent variable sequence are compressed into a real number with a global receptive field. This number represents the distribution of multivariate features along the temporal dimension and the global distribution of each variable sequence along the spatial dimension. The temporal squeeze process through GAP can be mathematically formulated as:

$$\tilde{x}_1 = \frac{1}{f} \sum_{i=1}^f x_1^i \quad (11)$$



Note: This plug-and-play module is designed to be inserted after the Bi-Mamba layer, enabling explicit modeling of feature importance in both temporal and spatial dimensions based on CD and CI strategies, respectively.

Fig. 4 The Architecture of the Proposed Multidimensional Dual-Path Attention Module STA

Similarly, the spatial squeeze process can be formulated as:

$$\tilde{x}_1 = \frac{1}{t} \sum_{i=1}^t x_i^i \quad (12)$$

The use of GAP in the squeeze processes ensures the lightweight characteristics of the STA module while still preserving the distribution of original features in the corresponding dimensions. The modeling process of the squeezed one-dimensional vector is achieved through the excitation operation composed of two fully connected layers. As shown in Fig. 4, the first fully connected layer (FC1) performs dimensionality reduction on the input, followed by a nonlinear transformation through the ReLU activation function (the reduction factor can be flexibly adjusted according to the hyperparameter r), while the second layer (FC2) restores the original dimensions and converts the output vector into normalized weights ranging from 0 to 1 using the Sigmoid function. This bottleneck structure further reduces the number of parameters, while the learned weights recalibrate the original feature maps through element-wise multiplication. The excitation operation can be expressed as follows:

$$\hat{x}_1 = \sigma(W_2 \delta(W_1 \tilde{x}_1)) \quad (13)$$

STA is also a plug-and-play module that can be inserted after each unidirectional branch of Bi-Mamba. It uses the CD and CI strategies to assign feature weights along the temporal and spatial dimensions, respectively. Before the outputs of the two branches are fused, STA retains more useful information from each branch and reduces the influence of less relevant features.

2.6 Model Design and Implementation Details

As a multi-lead-time forecasting model, ResBi-Mamba Plus is designed to forecast hourly runoff from 2 hours to 48 hours in the future. In terms of network architecture, ResBi-Mamba Plus supports the construction of dual residual block pathways with a depth of up to 6 layers, while in STA, the dimensionality reduction hyperparameter r is set to 4. Dropout regularization is also added after each unidirectional Mamba layer to prevent overfitting, with the dropout rate set to 0.2. The model utilizes the mean squared error (MSE) as the loss function, which is widely used in runoff forecasting. It penalizes the model by calculating the mean of the squared differences between the observed and predicted values, and its formula is:

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N (y_i^{\text{obs}} - y_i^{\text{pred}})^2 \quad (14)$$

where y_i^{obs} represents the observed value (label value), y_i^{pred} represents the predicted value by the model, and N denotes the number of samples. To

comprehensively and fairly evaluate models' performance, we employ the three most commonly used metrics in hydrology: mean absolute error (MAE), coefficient of determination (R^2), and Nash-Sutcliffe efficiency (NSE). The formulas for these metrics are as follows:

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |y_i^{\text{obs}} - y_i^{\text{pred}}| \quad (15)$$

$$\text{NSE} = 1 - \frac{\sum_{i=1}^N (y_i^{\text{obs}} - y_i^{\text{pred}})^2}{\sum_{i=1}^N (y_i^{\text{obs}} - \bar{y}^{\text{obs}})^2} \quad (16)$$

$$R^2 = \frac{\left[\sum_{i=1}^N (y_i^{\text{obs}} - \bar{y}^{\text{obs}})(y_i^{\text{pred}} - \bar{y}^{\text{pred}}) \right]^2}{\sum_{i=1}^N (y_i^{\text{obs}} - \bar{y}^{\text{obs}})^2 \sum_{i=1}^N (y_i^{\text{pred}} - \bar{y}^{\text{pred}})^2} \quad (17)$$

The study area of this paper is the Columbia River in Western North America, whose watershed spans major cities in both the United States and Canada, and is rich in ecosystems and significant socio-economic values. Owing to its diverse terrain conditions and climate patterns, the Columbia River Basin is representative and of broad research significance for runoff forecasting studies. Hourly river data are recorded by numerous water quality monitors situated throughout the watershed, and the data are compiled and published by Columbia Basin Research (CBR) [47], which provides sufficient open-access fine-grained hydrological data for this study. The dataset covers the period from January 1, 2016, to December 31, 2019, and contains 35,040 samples, with 75% used as the training set and 25% as the test set. One day of records (December 31, 2016) is missing due to a data export issue in the source dataset. Before model training and evaluation, all variables were preprocessed using min-max normalization. The model is trained for 100 epochs using the adaptive moment estimation (Adam) optimizer. Experiments are conducted on NVIDIA RTX 3070, based on PyTorch 1.12.1 + CUDA 12.8 environment. The runoff values are reported based on the source dataset's unit of kilocubic feet per second (kcfs), represented in this study as $10^3 \text{ ft}^3 \text{ s}^{-1}$ to ensure consistency with standard scientific notation. For international reference, 1 unit ($10^3 \text{ ft}^3 \text{ s}^{-1}$) is equivalent to approximately $28.317 \text{ m}^3 \text{ s}^{-1}$ in the International System of Units (SI).

3 Experiment Results and Discussion

3.1 Ablation Study

Compared to the baseline model Mamba-2, our

proposed ResBi-Mamba Plus introduces multiple principal enhancements, including the bidirectional architecture combined with the intra-block design that integrates dual path multidimensional STA attention. At the network level, it further improves information flow in deep layers through an additional auxiliary residual pathway and dense shortcuts. We conduct ablation experiments to examine the effectiveness of each component at different network depths. Robustness is further assessed by introducing different levels of perturbation into the input data. The forecasting horizon is fixed at 48 hours throughout these experiments. In this analysis, depth refers to the number of residual blocks rather than the total number of layers in the network.

As listed in Tables 1–3, bold numbers highlight the optimal prediction performance across all competing models within each experimental group. Table 1 shows the ablation results at different network depths. ResBi-Mamba Plus achieves lower errors than the other models in most settings, especially when the network becomes deeper. Mamba-2 and Bi-Mamba-STA present small improvements when the number of residual blocks is low. However, when the depth increases to six residual blocks, the errors of Mamba-2 and Bi-Mamba-STA increase by 6.5% and 5.3%, respectively. The relative increments of MAE are

Table 1 Performance Comparison at Different Depths

Depth	Model	MAE	NSE	R^2
1	Mamba-2	7.38	0.845	0.845
	Bi-Mamba-STA	7.02	0.856	0.856
	ResBi-Mamba Plus	7.04	0.858	0.858
2	Mamba-2	7.24	0.852	0.852
	Bi-Mamba-STA	7.08	0.857	0.857
	ResBi-Mamba Plus	6.79	0.868	0.868
4	Mamba-2	7.28	0.851	0.851
	Bi-Mamba-STA	7.13	0.854	0.854
	ResBi-Mamba Plus	6.53	0.878	0.878
6	Mamba-2	7.71	0.831	0.831
	Bi-Mamba-STA	7.39	0.845	0.845
	ResBi-Mamba Plus	6.22	0.887	0.887

Table 2 Robustness Evaluation at Lead Time 24 Under Varying Perturbation Levels

Perturbation Level	Model	MAE	NSE	R^2
Low	Mamba-2	6.59	0.876	0.876
	Bi-Mamba-STA	6.05	0.894	0.894
	ResBi-Mamba Plus	5.45	0.911	0.911
Medium	Mamba-2	7.11	0.860	0.860
	Bi-Mamba-STA	6.24	0.886	0.886
	ResBi-Mamba Plus	5.65	0.907	0.907
High	Mamba-2	8.12	0.826	0.826
	Bi-Mamba-STA	6.63	0.876	0.876
	ResBi-Mamba Plus	5.68	0.905	0.905

calculated by comparing the performance at depth six with the minimum MAE of each individual model across all tested depths. These results suggest that increasing the depth of Mamba-based models without an appropriate residual structure may lead to performance degradation. ResBi-Mamba Plus demonstrates a different trend. With the ResNet Plus architecture, its error decreases by approximately 11.6% as the depth increases. The result indicates that the proposed residual structure allows additional Bi-Mamba blocks to contribute to forecasting performance. Most existing Mamba-based models use only two or three Mamba blocks, whereas ResBi-Mamba Plus can be constructed with six blocks in this experiment. Since each Bi-Mamba block contains multiple internal layers for different operations, the six-block model has more than 100 hidden layers in total.

Moreover, Bi-Mamba-STA achieves lower errors than the baseline model at all depths. This result shows that adding the additional backward branch and the STA module improves forecasting performance. The backward branch allows the model to extract information from both directions of the historical sequence, while STA assigns weights to features in the temporal and spatial dimensions before the branch outputs are fused. Fig. 5 shows the training loss curves of the models with different components. ResBi-Mamba Plus converges faster than the other models. The faster convergence may be related to its residual structure. Dense shortcut connections between the two paths shorten the routes through which gradients are transmitted during training. Bi-Mamba-STA also converges faster than Mamba-2. This suggests that weighting the branch outputs before fusion may help the model make better use of the information extracted by the forward and backward branches.

Hydrological observations collected by automated monitoring stations may contain noise or missing values because of sensor errors, equipment maintenance, or extreme weather conditions. We therefore evaluate the models under controlled input perturbations, and three perturbation levels are considered. Gaussian noise with zero mean and standard deviations of 0.01, 0.05, and 0.1 is added, and 1%, 3%, and 5% of the input values are randomly masked. These settings correspond to the low, medium, and high perturbation levels, respectively. Since no standard perturbation setting is available for this evaluation, the three levels are used to represent increasing degrees of data corruption that may occur in hydrological monitoring. Table 2 reports the results. Bi-Mamba-STA obtains lower errors than Mamba-2 under all three perturbation levels, with a maximum MAE reduction of 18.3%. This result indicates that the bidirectional structure combined with STA is less sensitive to the tested disturbances than the unidirectional baseline.

Table 3 Performance Comparison at Different Lead Times

Lead Time	Model	MAE	NSE	R^2
2	TCN	3.41	0.957	0.957
	Bi-LSTM	3.35	0.956	0.956
	Transformer	4.62	0.937	0.937
	Informer	3.97	0.949	0.949
	iTransformer	3.56	0.947	0.947
	Mamba-2	4.12	0.941	0.941
	MambaFormer	3.61	0.953	0.953
	ResBi-Mamba Plus	3.15	0.961	0.961
4	TCN	4.62	0.927	0.927
	Bi-LSTM	4.66	0.932	0.932
	Transformer	5.05	0.921	0.921
	Informer	4.95	0.926	0.926
	iTransformer	4.42	0.932	0.932
	Mamba-2	4.63	0.931	0.931
	MambaFormer	4.58	0.934	0.934
	ResBi-Mamba Plus	4.11	0.943	0.943
8	TCN	5.88	0.895	0.895
	Bi-LSTM	5.57	0.910	0.910
	Transformer	5.81	0.902	0.902
	Informer	5.53	0.909	0.909
	iTransformer	5.43	0.902	0.902
	Mamba-2	5.57	0.907	0.907
	MambaFormer	5.70	0.902	0.902
	ResBi-Mamba Plus	4.67	0.930	0.930
12	TCN	5.79	0.894	0.894
	Bi-LSTM	5.83	0.903	0.903
	Transformer	6.49	0.881	0.881
	Informer	6.40	0.884	0.884
	iTransformer	5.33	0.904	0.904
	Mamba-2	5.66	0.906	0.906
	MambaFormer	6.49	0.881	0.881
	ResBi-Mamba Plus	5.22	0.918	0.918
24	TCN	6.75	0.868	0.868
	Bi-LSTM	6.44	0.883	0.883
	Transformer	7.25	0.856	0.856
	Informer	6.33	0.886	0.886
	iTransformer	5.69	0.900	0.900
	Mamba-2	6.34	0.886	0.886
	MambaFormer	7.00	0.860	0.860
	ResBi-Mamba Plus	5.32	0.915	0.915
48	TCN	7.12	0.848	0.848
	Bi-LSTM	7.15	0.852	0.852
	Transformer	8.92	0.782	0.782
	Informer	7.62	0.836	0.836
	iTransformer	6.90	0.856	0.856
	Mamba-2	7.24	0.852	0.852
	MambaFormer	8.53	0.797	0.797
	ResBi-Mamba Plus	6.22	0.887	0.887

ResBi-Mamba Plus achieves the lowest MAE in each setting. Compared with Mamba-2, its MAE is reduced by up to 30.0%, and its NSE remains above 0.9 under all three perturbation levels.

3.2 Comparative Experiments with Mainstream Models

We conduct comprehensive comparative experiments between the proposed ResBi-Mamba Plus and

three representative categories of mainstream models that are most widely used in the field of runoff forecasting, as introduced in Section 1. To evaluate the temporal robustness of the models across different forecasting horizons, we test them under six lead times ranging from short to long: 2, 4, 8, 12, 24, and 48 time steps. The models included in the comparison are as follows:

- 1) The TCN is a specialized CNN-based model

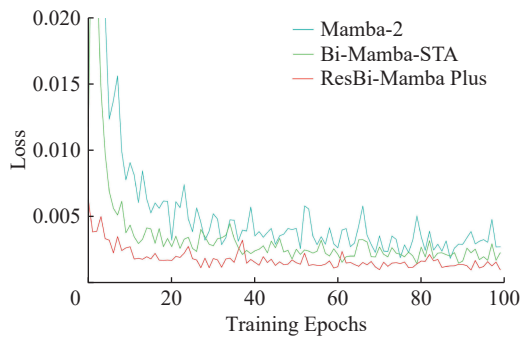


Fig. 5 Training Loss Curves of Different Models in the Ablation Study (depth = 6, forecasting horizon = 48)

for time series forecasting, which uses modified causal convolutions and dilated convolutions to prevent information leakage and effectively enlarge the receptive field.

2) The Bi-LSTM adopts a novel bidirectional recurrent architecture and is regarded as one of the most advanced and high-performing variants of LSTM.

3) The Transformer replaces traditional recurrent architectures with the self-attention mechanism and has become one of the most widely used models in deep learning.

4) The Informer builds upon the traditional Transformer by incorporating the ProbSparse self-attention mechanism and the self-attention distilling strategy to reduce computational complexity and parameter count. It is considered one of the SOTA time series forecasting models.

5) The iTransformer adopts the inverted modeling method based on the CI strategy, treating variables as the primary dimension to learn feature representations of multivariate time series. It stands as one of the state-of-the-art (SOTA) models for long-term forecasting tasks.

6) The Mamba-2 is the latest paradigm within the SSM family and also serves as the backbone of our proposed methodology.

7) The MambaFormer is a popular hybrid model in time series forecasting [48], combining the respective strengths of the Mamba and Transformer architectures. It continues to evolve with the emergence of new variants.

Runoff forecasting studies have gradually moved from single-step to multi-step. It is therefore necessary to examine model performance at different lead times. Table 3 and Fig. 6 report the comparative results. At short and medium lead times, traditional forecasting models perform similarly to several Transformer- and Mamba-based models. Bi-LSTM also obtains good results at several lead times. Since Bi-LSTM uses a bidirectional structure, its performance suggests that processing the historical sequence in both directions can be useful for runoff forecasting. As the lead time increases, iTransformer and Mamba-2, both representing the latest SOTA models within their respective branches, demonstrate

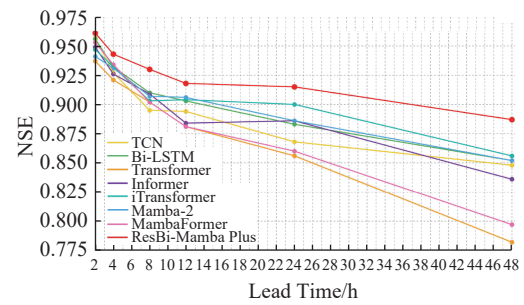


Fig. 6 NSE of Mainstream Time Series Forecasting Models at Various Lead Times

remarkable temporal robustness in long-term forecasting. The CI-based iTransformer demonstrates substantial performance improvement over the vanilla Transformer through capturing intra-channel dependencies, establishing itself as the second-best model across several lead times. As shown in Fig. 7, we further conduct a visual comparison of the actual (observed) and predicted runoff curves for the long-term forecasting scenario (lead time = 24) in different seasons. The results show that both iTransformer and ResBi-Mamba Plus, which achieve SOTA performance, are able to closely fit the actual curves. In contrast, the vanilla Transformer suffers from its quadratic complexity, where the growth of the attention matrix in longer sequences impedes effective training. Consequently, even the hybrid MambaFormer, which integrates Mamba-2 and Transformer components, still underperforms in long-term horizons. The results clearly show that across all six evaluated lead times, our proposed ResBi-Mamba Plus consistently outperforms all competing models in terms of various evaluation metrics, highlighting its superior overall performance. At the longest forecasting horizon (lead time = 48), ResBi-Mamba Plus achieves a 9.9% reduction in error compared to iTransformer, the widely recognized SOTA model for long-term forecasting. This demonstrates that our proposed methodology based on both CD and CI strategies is more suitable for long-term, fine-grained runoff forecasting than single-strategy models like iTransformer. Furthermore, the performance advantage of ResBi-Mamba Plus is even more pronounced when compared to other mainstream models: it achieves 30.3% and 18.4% lower errors than the Transformer and the Informer, respectively, strongly validating the effectiveness and advancement of our proposed framework.

3.3 Interpretability of Attention Module

To further examine how STA assigns importance to different time steps, we use an interpretable machine learning (IML) method to visualize its temporal attention weights. Fig. 8 presents the heatmaps of the forward and backward branches of Bi-Mamba. Brighter colors indicate larger weights at the corresponding time steps. In the forward branch,

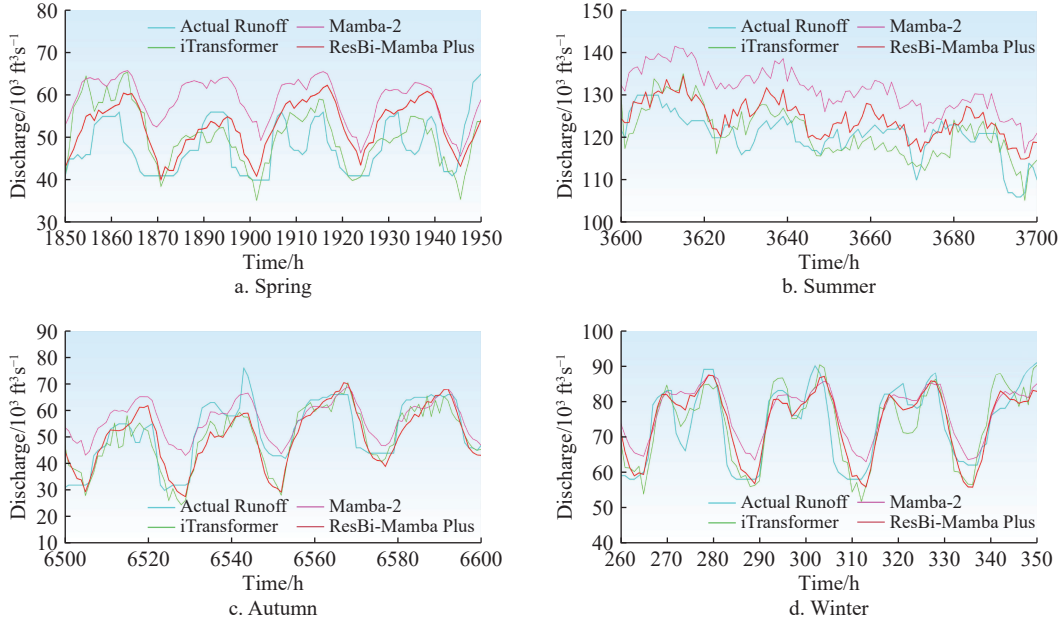


Fig. 7 Comparison of Actual and Predicted Runoff Curves in Different Seasons

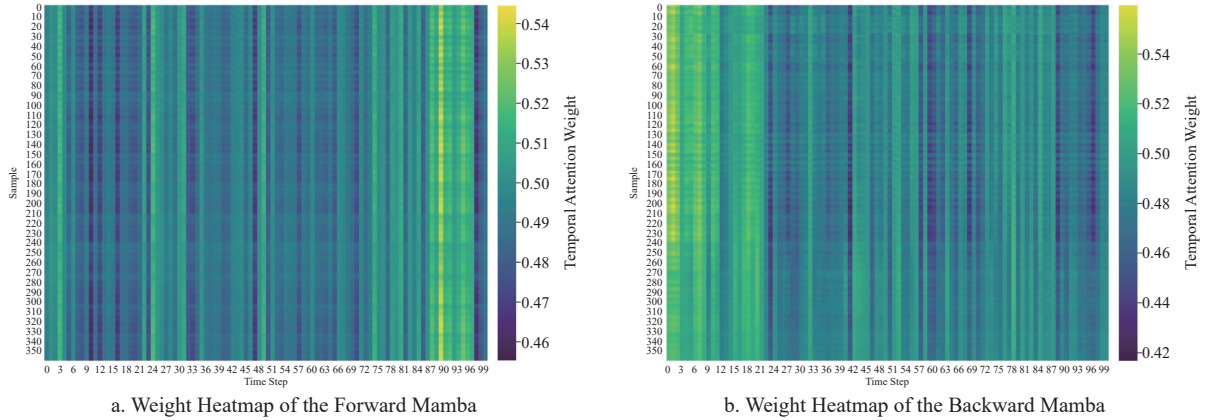


Fig. 8 Temporal Attention Weight Heatmaps of Forward and Backward Bi-Mamba Branches

higher weights are mainly distributed on the right side of the heatmap, where the observations are closer to the forecasting horizon. The backward branch processes the same input sequence in reverse order. Therefore, observations close to the forecasting horizon appear on the left side of its heatmap and also receive higher weights. These results are consistent with a common property of time series data: observations closer to the forecasting point are generally more closely related to the current state of the sequence. STA captures this local temporal information in both branches. At the same time, some time steps farther from the forecasting horizon also receive relatively high weights. This suggests that the model does not rely only on recent observations, but also uses information from earlier time steps when it is useful for forecasting. This pattern is consistent with the presence of long-term dependencies in the runoff sequence.

4 Conclusion

This paper presents ResBi-Mamba Plus for

interpretable multi-lead-time hourly runoff forecasting. The model uses two parallel residual block paths with dense shortcut connections to improve information transmission in the deep layers. Bi-Mamba is proposed as the backbone to process historical sequences in both forward and backward directions. An STA module is further inserted after each unidirectional Mamba branch. It uses the CD and CI strategies to assign feature weights along the temporal and spatial dimensions, respectively. This design helps the model use relevant information from both branches before their outputs are fused. Ablation results prove the effectiveness of the proposed components, while the comparative results indicate that ResBi-Mamba Plus achieves lower forecasting errors than the mainstream models at different lead times. The proposed framework provides a new way to use Mamba-based models for multivariate hourly runoff forecasting.

However, this paper still has two main limitations. First, the experiments consider forecasting horizons of up to 48 hours. The performance of the proposed

model at longer horizons has not yet been examined. Second, the current model is mainly data-driven and does not explicitly include additional hydrometeorological variables or physical constraints. These factors may be useful for improving the reliability and interpretation of forecasts under complex conditions. Future work will examine longer forecasting horizons and extend the model by incorporating additional information and physical knowledge. The application of Mamba-based structures to other time series forecasting tasks will also be investigated.

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